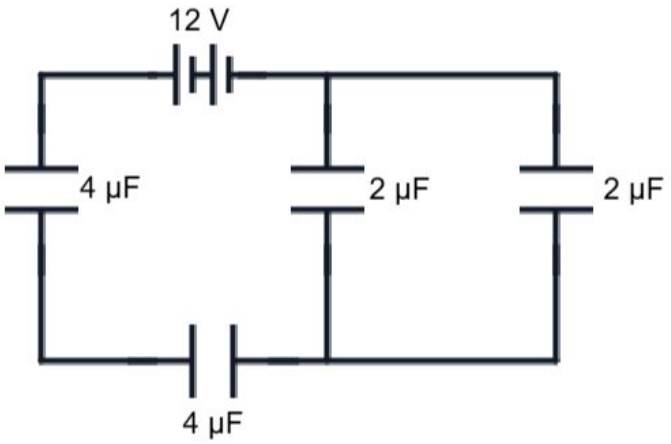
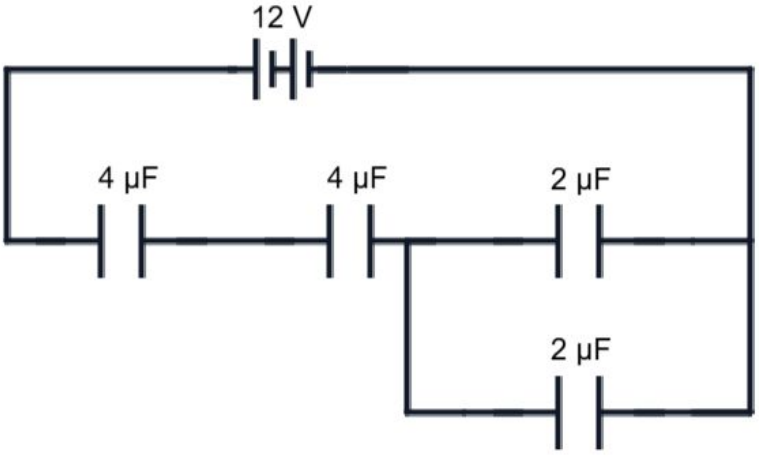
23	The diagram shows a network of capacitors connected to a 12 V battery of negligible internal resistance.



What is the total amount of charge circulated by the 12 V battery?

	A	16 μC	**B**	24 μC	**C**	36 μC	**D**	110 μC

L1 **Answer: A**



$C_{2\mu F \text{ resistors in parallel}}$ = 2 + 2 = 4 μF

$C_{eq} = \left(\frac{1}{4} - \frac{1}{4} + \frac{1}{4}\right)^{-1}$ = 1.3333 μF

Total charge circulated = $C_{eq} V_{eq}$ = (1.3333 μF) (12 V) = 16 μC

OR

p.d. across each 4 μF capacitor = p.d. across the two parallel 2 μF capacitors
= 12 V / 3 = 4 V

Total charge circulated = charge stored on each of the 4 μF capacitor = total charge stored on the two parallel 2 μF capacitors
= (4 μF)(4 V) = 16 μC